

COURSE 2A

C2-01 ADVANCED EXCEL & POWER QUERY

BEGINNER-FIRST TEACHING BOOK - RC1

Senior / BI Analyst optional depth path

Formula architecture → Power Query → refreshable reporting

THIS BOOK IS YOUR TEACHER

Rebuilt from the frozen Course 2A baseline; technically useful content preserved only after audit.

This book is your teacher

How to use C2-01 without becoming dependent on copied formulas

THE LEARNING LOOP

Read the mental model first. Watch only the assigned visual segment if it helps. Follow one worked example. Then build a different task in the workbook. Save your genuine attempt, validate it, explain it back, and only then open the protected review. Close review before the changed retry.

MODERN EXCEL COMPATIBILITY

Preferred hands-on runtime: current Microsoft 365 / Excel 2024 on Windows. XLOOKUP and dynamic arrays may not work in older Excel versions. If a feature is unavailable, use the taught fallback for the concept, but do not claim hands-on mastery of a feature your runtime cannot execute.

POWER QUERY RUNTIME

Power Query Desktop is required for AX6–AX10 hands-on mastery. Excel for Windows is the intended route; Power BI Desktop can reinforce many Power Query transformations but does not make an Excel-specific workflow test identical.

WHAT 'ADVANCED' MEANS HERE

Not longer formulas. Advanced means: clear business contracts, repeatable transformation, controlled source scope, key/grain checks, independent reconciliation, refresh/failure recovery and an artifact another analyst can inherit.

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Prerequisite and tool map

What Course 1/C2-00 knowledge is assumed and what C2-01 adds

ASSUMED

Course 1: formulas, basic lookup awareness, Excel Tables, filters, basic PivotTables, basic Power Query exposure, simple data cleaning and reconciliation. C2-00: metric contracts, grain, source-of-truth and validation language. If a Course 1 skill is weak, repair that skill rather than skipping the senior control layer.

TOOLS / FILES

Current Excel for Windows preferred. Teaching Book + Practice Workbook + practice CSV folder + Mini-Lab + protected review + Gate A. B/C stay protected. Keep a clean copy of raw files before transformations.

NEW DEPTH

Formula architecture; key-safe lookup design; spill-safe arrays; auditable KPI grids; Pivot refresh/reconciliation; Power Query architecture; folder ingestion; merge/append grain safety; parameters; operational month-end release.

DIRECT DA → DE POSITIONING

C2-01 is optional Senior/BI depth. Some concepts overlap C2B—especially Power Query-style ingestion thinking, grain, validation and repeatability—but C2A itself must not become a hidden prerequisite. The direct route remains Course 1 → C2B → C3 when C2B entry competencies are met.

Practice, review and Gate route

Attempt first; changed retry; competency Gate



LESSON MASTERY

A formula/query that runs is not enough. You must predict the result shape, validate row/key/file contribution or totals, explain why the result is safe, save the genuine attempt, then reproduce the skill on changed data after review is closed.

MINI-LAB

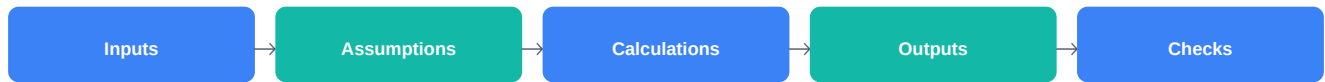
After AX1–AX10, run a month-end refresh case. The Mini-Lab deliberately includes an unexpected backup file and a schema-drift file. You must block release, diagnose, repair and rerun controls.

GATES

Gate A is independent. Weak skill → targeted repair → changed evidence. Gate B/C are fresh full reassessments only when explicitly assigned. A passing percentage cannot override an unresolved critical source/key/grain/reconciliation failure.

AX1 - Advanced formulas without formula spaghetti

Page A - why, outcome, prerequisite, mental model and vocabulary



WHY IT MATTERS

A workbook becomes risky when business rules are copied into many long formulas. One rule change can leave half the workbook using the old value and half using the new value.

WHAT YOU LEARN

Build a layered workbook: raw inputs, visible assumptions, calculations, outputs and controls. Use structured references or named cells so business constants live in one place.

PREREQUISITE

Course 1 formulas, cell references, Excel Tables, IF/SUM basics. If structured references are unfamiliar, read the example slowly; they are defined here.

NEW WORDS BEFORE WE USE THEM

Structured reference: A formula reference that uses Excel Table and column names, such as Sales[Amount], instead of fixed coordinates. Assumption cell: One visible cell that stores a business constant such as a tax rate, target, or threshold. Control formula: A formula designed to expose an error or mismatch rather than produce the business output.

AX1 - Follow with me once

Page B - Net revenue with one controlled discount rule

Net revenue with one controlled discount rule

- 1 Create an Excel Table named tSales with columns OrderID, Gross, DiscountRate and Net.
- 2 Put the default discount in a visible assumptions cell named pDefaultDiscount instead of typing 0.05 into many formulas.
- 3 Use a readable formula pattern. Example: `=[@Gross]*(1-IF([@DiscountRate]="",pDefaultDiscount,[@DiscountRate]))`.
- 4 Create a summary output with `=SUM(tSales[Net])`. Then create a separate control comparing that total to an independently calculated gross-minus-discount total.
- 5 Add an invariant check: no row may have Net > Gross. A formula can calculate `=COUNTIFS(tSales[Net], ">"&tSales[Gross])` only in versions that support this exact pattern reliably; otherwise use a helper flag and COUNTIF. The principle is the control, not memorizing one clever formula.

EXPECTED RESULT

Business constants are visible, formulas are shorter, and a reviewer can trace input → rule → calculation → output → check.

WHY IT WORKED

One assumption is changed once and flows through the calculation. The control is separate from the output, so it can fail visibly.

AX1 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: create a shipping-cost workbook where FuelSurcharge and FreeShippingThreshold are visible assumptions. Produce line cost, order total and two controls.

COMMON BEGINNER MISTAKES

1. Hard-coding the same percentage in many formulas. | 2. Building one giant nested formula because it looks advanced. | 3. Mixing raw data and manual overrides inside the calculation column. | 4. Checking only that the formula returns a number.

STUCK? RECOVERY

Split the formula into pieces. First verify the row input, then the assumption cell, then the calculation, then the summary, then the independent control. Do not debug all layers at once. If still stuck, use only the partial skeleton in the practice workbook; save the attempt before opening review.

VALIDATE

Change the surcharge once and confirm only intended outputs change. Reconcile line totals to the summary and test one impossible-value rule.

AX1 - Prove mastery

Page D - professional version, teach-back, review/retry and evidence

THEN SHOW THE PROFESSIONAL VERSION

Professional Excel files make business rules discoverable. Use Tables, named assumptions, clear calculation columns, comments/notes where needed, and controls that a future owner can rerun.

TEACH IT BACK

Explain why fewer formulas can be more professional than a more complicated formula.

PROTECTED REVIEW + CHANGED RETRY

Save your independent workbook first. Then open AX1 review, close it, and rebuild with a changed fee rule without copying the finished formula.

EVIDENCE / MASTERY

Workbook screenshot/file + assumption block + two controls + 60–90 second explain-back.

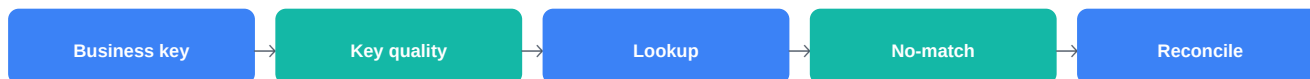
RECOMMENDED VIDEO: Best Excel Formulas in an hour - Beginner to PRO Masterclass

1:51–8:28 for conditional summaries; 1:01:41 for report-with-formulas context.

https://www.youtube.com/watch?v=dVJFyqI_Nw4

AX2 - XLOOKUP, INDEX/MATCH and lookup design

Page A - why, outcome, prerequisite, mental model and vocabulary



WHY IT MATTERS

A lookup can return a plausible value while the lookup table contains duplicate keys, missing keys, or the wrong business identifier. The formula may be syntactically correct and still be analytically unsafe.

WHAT YOU LEARN

Design a lookup contract: business key, lookup range, return field, exact-match behavior, no-match handling, duplicate-key test and unmatched-key reconciliation. Use INDEX/MATCH as a transferable fallback mental model.

PREREQUISITE

AX1 formula architecture; Course 1 exact-match lookup awareness. Modern XLOOKUP requires a supported Excel version.

NEW WORDS BEFORE WE USE THEM

Business key: A column or column combination intended to identify the business entity used for matching. **Exact match:** Return only when the lookup key equals the target key according to the chosen matching rule. **Unmatched key:** A source key for which the lookup table has no valid matching record.

AX2 - Follow with me once

Page B - Attach product category to order lines safely

Attach product category to order lines safely

- 1
- 2
- 3
- 4
- 5

Before writing XLOOKUP, test the product master: ProductID should be unique and nonblank.

Use visible no-match output: =XLOOKUP([@ProductID],tProduct[ProductID],tProduct[Category],"UNMATCHED").

Count UNMATCHED results before replacing or hiding them. Investigate whether the source key is new, mistyped, or missing from the master.

Fallback mental model: MATCH finds the row position of the key; INDEX returns a value from the return column at that position. The business contract is the same even when the function changes.

Reconcile row count: a lookup should enrich each source row; it should not silently add or remove source rows.

EXPECTED RESULT

Every source row has one trusted category or a visible UNMATCHED exception, and duplicate lookup keys are investigated rather than ignored.

WHY IT WORKED

The key-quality check happens before the retrieval, so a successful formula cannot hide a broken master table.

AX2 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: attach ServiceTier from an Account master to ticket rows. The dirty master contains one duplicated AccountID and two source IDs not in the master. Build visible exception handling.

COMMON BEGINNER MISTAKES

1. Wrapping IFERROR around everything before inspecting the cause. | 2. Using approximate match accidentally. | 3. Assuming a code is unique because it looks like an ID. | 4. Choosing XLOOKUP only because it is newer, not because the contract is clear.

STUCK? RECOVERY

Run three checks in order: blank key count, duplicate key count in master, unmatched source key count. Repair the data contract before changing the formula. If still stuck, use only the partial skeleton in the practice workbook; save the attempt before opening review.

VALIDATE

Source row count unchanged; master business key unique after repair; unmatched count explained; spot-check at least three IDs.

AX2 - Prove mastery

Page D - professional version, teach-back, review/retry and evidence

THEN SHOW THE PROFESSIONAL VERSION

Use lookup formulas for appropriate spreadsheet-scale enrichment, but preserve key-quality and unmatched controls. For repeatable larger transformations, the same reasoning transfers to Power Query merges and database joins.

TEACH IT BACK

Explain why XLOOKUP does not protect you from a bad business key.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. After review, use a changed employee-to-department lookup with different no-match behavior.

EVIDENCE / MASTERY

Lookup contract + dirty-key check + unmatched list + reconciled row count + explain-back.

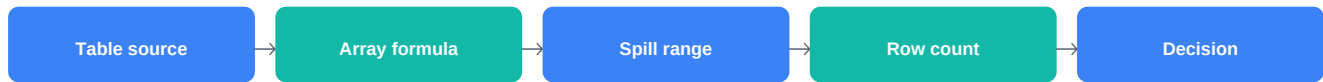
REQUIRED VISUAL SUPPORT: 10 Advanced XLOOKUP Tips & Tricks

1:15–4:18 for basic XLOOKUP and wildcard behavior; use later tricks only as enrichment.

<https://www.youtube.com/watch?v=YyhJe5tMq58>

AX3 - Dynamic arrays: FILTER, UNIQUE, SORT and spill-safe workflows

Page A - why, outcome, prerequisite, mental model and vocabulary



WHY IT MATTERS

Dynamic arrays make analyst views fast, but a blocked spill range or unsupported workbook version can make a correct-looking design fail. You must understand the returned array, not just copy a formula.

WHAT YOU LEARN

Use FILTER, UNIQUE and SORT to create live analysis views; understand spill ranges, #SPILL!, source expansion and version limits; validate returned row counts independently.

PREREQUISITE

AX1 Tables and formulas. Requires a modern Excel version supporting dynamic arrays for hands-on mastery.

NEW WORDS BEFORE WE USE THEM

Dynamic array: A formula result that can return multiple cells from one formula. Spill range: The cells automatically occupied by a dynamic-array result. #SPILL!: An error shown when Excel cannot place the full dynamic-array result into its intended cells.

AX3 - Follow with me once

Page B - Live exception list by region

Live exception list by region



With tSales as the source, create =SORT(UNIQUE(tSales[Region])) to generate a live region list.

Create a separate view: =FILTER(tSales[[OrderID]:[Net]],tSales[Status]="Open","No open orders").

Predict the expected row count using COUNTIF before trusting the spill.

Type a value into one cell where the result needs to spill. Observe #SPILL!, inspect the obstruction, remove it, and rerun the row-count control.

Add a new row to the source Table and confirm the dynamic result expands automatically.

EXPECTED RESULT

A live spill-safe analysis view that expands with the source and has an independent count control.

WHY IT WORKED

The Table handles source expansion; the dynamic-array formula handles output expansion; the independent count checks completeness.

AX3 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: build a sorted unique owner list and a live FILTER view of unresolved priority tickets. Prove count parity.

COMMON BEGINNER MISTAKES

1. Putting spill formulas inside a Table calculated column. | 2. Leaving manual notes inside a spill range. | 3. Ignoring workbook/version compatibility. | 4. Assuming a visually correct list contains every eligible row.

STUCK? RECOVERY

Select the formula cell, inspect the intended spill border/error, clear obstructions, verify the source Table range, then compare the returned count to COUNTIF/COUNTIFS. If still stuck, use only the partial skeleton in the practice workbook; save the attempt before opening review.

VALIDATE

Independent eligible-row count equals the spill output count; source expansion works; no manual data sits inside spill range.

AX3 - Prove mastery

Page D - professional version, teach-back, review/retry and evidence

THEN SHOW THE PROFESSIONAL VERSION

Dynamic arrays are excellent for QA lists, lightweight reporting and helper views. For governed repeatable transformations, decide whether Power Query or the data model is the more maintainable layer.

TEACH IT BACK

Explain the difference between one formula cell and its spilled result.

PROTECTED REVIEW + CHANGED RETRY

Save attempt, then review. Changed retry: a product/region exception list with a different condition and sort order.

EVIDENCE / MASTERY

Working dynamic-array view + row-count control + #SPILL! recovery evidence + explain-back.

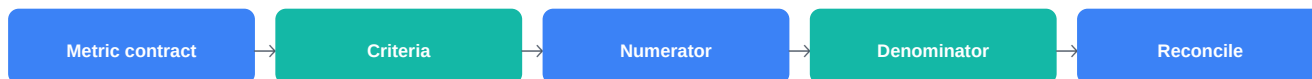
RECOMMENDED VIDEO: Best Excel Formulas in an hour - Beginner to PRO Masterclass

8:28–12:21 FILTER; 33:44 UNIQUE; 57:22 FILTER + dates.

https://www.youtube.com/watch?v=dVJFyqI_Nw4

AX4 - SUMIFS, COUNTIFS and auditable conditional KPIs

Page A - why, outcome, prerequisite, mental model and vocabulary



WHY IT MATTERS

Conditional formulas are easy to write and easy to mis-scope. One wrong date boundary or denominator can produce a convincing KPI that answers a different question.

WHAT YOU LEARN

Translate a metric contract into SUMIFS/COUNTIFS criteria; separate numerator and denominator; use real dates; reconcile conditional summaries to independent totals.

PREREQUISITE

SAF4 metric contracts from C2-00 and AX1 formula architecture.

NEW WORDS BEFORE WE USE THEM

Criteria range: The range evaluated against a condition in SUMIFS/COUNTIFS. Numerator: The count or amount on top of a rate. Denominator: The eligible population underneath a rate.

AX4 - Follow with me once

Page B - Completed revenue and completion rate by region/month

Completed revenue and completion rate by region/month

1

Define period first: use start date $\geq$ first day and date $<$ first day of next month to avoid time-of-day edge cases.

2

Completed revenue pattern:

`=SUMIFS(tSales[Net],tSales[Region],$A2,tSales[Status],"Completed",tSales[OrderDate], ">="&B$1,tSales[OrderDate], "<"&EDATE(B$1,1`

3

Build numerator and denominator in separate cells before dividing. Example numerator = completed orders; denominator = eligible non-test orders.

4

Use `=IFERROR(Numerator/Denominator,"")` only after denominator quality is understood; do not use IFERROR to hide a zero-population definition problem.

5

Reconcile all region totals back to an independently filtered/pivoted source total.

EXPECTED RESULT

A KPI grid with explicit criteria, visible numerator/denominator and a grand-total reconciliation.

WHY IT WORKED

The metric definition is converted into auditable criteria one field at a time, and the period uses an exclusive upper boundary.

AX4 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: build On-Time Delivery Rate by region for three months with visible numerator/denominator and a control total.

COMMON BEGINNER MISTAKES

1. Using text months instead of real dates. | 2. Forgetting one criterion on the denominator. | 3. Using SUMIF/SUMIFS totals as their own validation. | 4. Hiding zero denominators without explaining why they are zero.

STUCK? RECOVERY

Write the KPI in English. Highlight population, event/status, date basis and segment. Map each phrase to exactly one formula criterion, then test criteria one at a time. If still stuck, use only the partial skeleton in the practice workbook; save the attempt before opening review.

VALIDATE

Manually filter one region/month, compare numerator and denominator, then reconcile grid totals to source.

AX4 - Prove mastery

Page D - professional version, teach-back, review/retry and evidence

THEN SHOW THE PROFESSIONAL VERSION

Keep metric logic consistent across spreadsheet, Power BI and SQL layers. A professional handoff includes the metric contract next to the formula design.

TEACH IT BACK

Explain why a rate is usually harder to define than a simple count.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry uses refund rate with a different eligible population.

EVIDENCE / MASTERY

KPI contract + conditional formulas + independent reconciliation + explain-back.

RECOMMENDED VIDEO: Best Excel Formulas in an hour - Beginner to PRO Masterclass

1:51–8:28 SUMIFS/COUNTIFS/AVERAGEIFS; 1:01:41 report-with-formulas.

https://www.youtube.com/watch?v=dVJFyqI_Nw4

AX5 - Advanced PivotTables and business summaries

Page A - why, outcome, prerequisite, mental model and vocabulary



WHY IT MATTERS

PivotTables can answer business questions quickly, but they will happily count text when you expected a sum, summarize duplicated data, or stay stale when the source changes.

WHAT YOU LEARN

Create a PivotTable from a clean Table; choose Rows/Columns/Values/Filters intentionally; change aggregation; group dates; use slicers; refresh and reconcile grand totals.

PREREQUISITE

Course 1 basic PivotTable exposure, AX4 metric definitions, clean tabular source.

NEW WORDS BEFORE WE USE THEM

Values aggregation: The summary operation applied in the Values area, such as Sum, Count or Average. Slicer: A clickable visual filter connected to a PivotTable. Pivot cache: Excel's internal stored representation used by PivotTables; refresh is needed when source data changes.

AX5 - Follow with me once

Page B - Service calls by customer and satisfaction

Service calls by customer and satisfaction



Start from one Excel Table with one header row and no subtotal rows.

Build a PivotTable: Customer in Rows; CallID in Values. Confirm whether Excel chose Count or Sum and why.

Add Satisfaction to Columns or Values depending on the question; do not treat categories as numeric measures.

Add Date and group to month only after confirming the source column contains real dates.

Record the source control total and compare it to Pivot grand total. Add one new source row, refresh, and prove the Pivot changed.

EXPECTED RESULT

A PivotTable whose field choices answer a stated question, uses the correct aggregation, refreshes, and reconciles to source.

WHY IT WORKED

The question drives the field layout; the source Table controls expansion; the reconciliation catches stale or duplicated inputs.

AX5 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: analyze support tickets by team, month and priority. Create one slicer and one trend PivotChart. Reconcile total ticket count.

COMMON BEGINNER MISTAKES

1. Dragging fields until the table looks interesting without stating the question. | 2. Trusting Excel's default Sum/Count. | 3. Building pivots from ranges that exclude new rows. | 4. Using a PivotTable to hide source data-quality problems.

STUCK? RECOVERY

Return to the source Table. Confirm grain, field data types and control total. Rebuild the smallest pivot with one Row and one Value, verify aggregation, then add complexity. If still stuck, use only the partial skeleton in the practice workbook; save the attempt before opening review.

VALIDATE

Source row count/grain documented; Pivot grand total reconciles; refresh test changes as expected; aggregation rationale saved.

AX5 - Prove mastery

Page D - professional version, teach-back, review/retry and evidence

THEN SHOW THE PROFESSIONAL VERSION

Use PivotTables for rapid exploration, QA and stakeholder working sessions. Move stable repeatable transformations upstream and document any calculated fields/measures.

TEACH IT BACK

Explain why a PivotTable is a question-answering layer, not a data-cleaning layer.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry uses orders by category and channel with a different value aggregation.

EVIDENCE / MASTERY

Pivot file + business question + aggregation rationale + refresh proof + reconciliation.

REQUIRED VISUAL SUPPORT: Excel Pivot Tables - Master Data Analysis in just 45 minutes

0:51–24:40 required core; 31:43+ optional enrichment.

<https://www.youtube.com/watch?v=zuSNd1ZMfBI>

AX6 - Power Query architecture and repeatable imports

Page A - why, outcome, prerequisite, mental model and vocabulary



WHY IT MATTERS

Manual copy/paste cleaning must be repeated every month and is hard to audit. Power Query stores a repeatable sequence of transformations—but a successful refresh still needs controls.

WHAT YOU LEARN

Understand the Power Query Editor, queries, Applied Steps, source, types, transformations, load destination and refresh. Build steps that are readable and predictable.

PREREQUISITE

Basic Excel tables/files. No M language knowledge is required; M is introduced only as the code behind Applied Steps.

NEW WORDS BEFORE WE USE THEM

Query: A repeatable set of instructions that connects to and transforms data. Applied Step: One recorded transformation in a Power Query query. M: The Power Query formula language generated behind many UI actions. Load: Sending the query result to a worksheet, Data Model or another supported destination.

AX6 - Follow with me once

Page B - Import and clean one monthly CSV

Import and clean one monthly CSV



1 Use Data → Get Data → From Text/CSV (exact ribbon wording can vary by Excel build), then choose Transform Data.

2 In Power Query Editor, identify Queries pane, data preview, Applied Steps and formula bar. Rename the query qSales_Staging.

3 Set OrderDate to Date and Amount to Decimal Number deliberately. Trim text status values and rename unclear columns.

4 Before filtering, record source row count/control total. Apply a filter and state why row count should change.

5 Rename important steps so a future analyst can read the pipeline. Load to a worksheet Table only after validation.

EXPECTED RESULT

A refreshable query with readable steps, explicit data types, known row-count changes and a documented output.

WHY IT WORKED

Transformation logic is stored once and reruns when the source changes; controls make refresh success different from data correctness.

AX6 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: import a customer CSV, standardize dates/text, create one conditional column, and save before/after controls.

COMMON BEGINNER MISTAKES

1. Cleaning manually before Power Query and losing reproducibility. | 2. Accepting automatically inferred types without checking. | 3. Deleting error rows before investigating why they are errors. | 4. Using dozens of anonymous steps with no logic grouping.

STUCK? RECOVERY

Select Applied Steps from top to bottom. Find the first step where row count/type/value becomes wrong. Fix or move that step; do not restart the entire query blindly. If still stuck, use only the partial skeleton in the practice workbook; save the attempt before opening review.

VALIDATE

Source row count known; expected row-changing steps explained; types correct; refresh reproduces same result.

AX6 - Prove mastery

Page D - professional version, teach-back, review/retry and evidence

THEN SHOW THE PROFESSIONAL VERSION

Organize queries by stage (staging/reference/output), filter early when appropriate, postpone expensive operations, set types deliberately, and document complex steps.

TEACH IT BACK

Explain Power Query as instructions, not as a cleaned copy of data.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry uses a service log with a different date/text problem.

EVIDENCE / MASTERY

Excel file with query + Applied Steps screenshot + row/total controls + explain-back.

REQUIRED VIDEO: Power Query - Beginner to PRO Masterclass in 30 minutes

4:54–13:36 editor/clean/load; 17:50–28:25 business cleaning; 33:35–35:17 load/refresh.

<https://www.youtube.com/watch?v=MMdcczmULrU>

AX7 - Cleaning and transforming multiple files

Page A - why, outcome, prerequisite, mental model and vocabulary



WHY IT MATTERS

Folder combine can automate monthly ingestion, but it can also silently ingest backup files, omit a month, or break when one file's schema drifts.

WHAT YOU LEARN

Use From Folder; filter intended files before combining; understand sample-file transformation; preserve Source.Name; validate file count, row contribution, schema and totals; diagnose drift.

PREREQUISITE

AX6 single-query workflow. Comfortable with CSV files and expected column names.

NEW WORDS BEFORE WE USE THEM

Folder connector: A Power Query connector that lists files from a folder or file-system-like source. Sample file: The representative file Power Query uses to define the transformation applied to combined files. Schema drift: A change to expected columns/types/structure between source files. File contribution: The rows or totals contributed by each source file to the combined output.

AX7 - Follow with me once

Page B - Combine January–March sales safely

Combine January–March sales safely



1 Point From Folder to a dedicated practice folder. Inspect the file list before pressing Combine.

2 Filter Extension = .csv and explicitly exclude names containing backup/IGNORE. Do this before combine.

3 Combine only intended monthly files. Keep Source.Name in the output so every row retains file lineage.

4 Build a contribution table: Source.Name, row count and Net total. Expect exactly three clean monthly files.

5 Add the provided April drift file only after the clean run. Diagnose the changed column/type instead of deleting the failing file.

EXPECTED RESULT

One combined table with source lineage and an auditable per-file contribution check; drift creates a visible exception, not silent omission.

WHY IT WORKED

File scope is controlled before combine and Source.Name remains available for reconciliation after combine.

AX7 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: combine three service CSVs, deliberately introduce one backup and one drift file, then create file-count and contribution controls.

COMMON BEGINNER MISTAKES

1. Using Combine while backup files are in the same folder. | 2. Deleting Source.Name too early. | 3. Fixing one monthly file manually outside the pipeline. | 4. Treating a successful refresh as proof that all expected months arrived.

STUCK? RECOVERY

Return to the folder listing query. Compare actual filenames to expected filenames, then inspect sample transformation and the first failing file/column. Repair the structural cause. If still stuck, use only the partial skeleton in the practice workbook; save the attempt before opening review.

VALIDATE

Expected vs actual file list; row counts by file; total by file; schema exception documented; no backup contribution.

AX7 - Prove mastery

Page D - professional version, teach-back, review/retry and evidence

THEN SHOW THE PROFESSIONAL VERSION

Production folder ingestion should define filename/scope rules, expected cadence, schema contract, file contribution controls and failure handling before refresh is considered releasable.

TEACH IT BACK

Explain why Source.Name is a data-quality field during ingestion.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry uses a different folder with one missing expected month and one unexpected extra file.

EVIDENCE / MASTERY

Folder query + expected/actual file list + per-file contributions + drift recovery + explain-back.

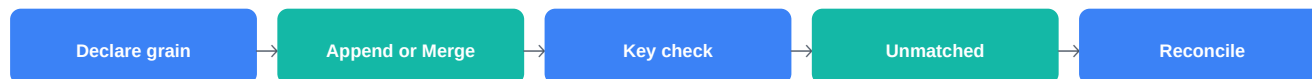
REQUIRED VIDEO: Power Query - Beginner to PRO Masterclass in 30 minutes

Use 4:54–13:36 for editor flow; course lab teaches folder-specific controls.

<https://www.youtube.com/watch?v=MMdcczmULrU>

AX8 - Merge vs append and relationship/grain mistakes

Page A - why, outcome, prerequisite, mental model and vocabulary



WHY IT MATTERS

Append and merge solve different problems. Choosing the wrong one—or expanding a merge against duplicate dimension keys—can multiply records and totals without an obvious error.

WHAT YOU LEARN

Append compatible rows; merge columns by keys; declare input/output grain; verify right-side key uniqueness; use join kinds deliberately; reconcile rows/totals before and after.

PREREQUISITE

SAF5 grain reasoning, AX6 Power Query basics, AX7 file lineage.

NEW WORDS BEFORE WE USE THEM

Append: Stack the rows of tables into one combined table by column names. Merge: Join columns from two queries using matching key values. Join kind: Rule defining which matched/unmatched rows are retained, such as Left Outer or Inner. Left anti: Rows from the left table that have no match in the right table.

AX8 - Follow with me once

Page B - Append periods, then enrich with product master

Append periods, then enrich with product master

- 1 Declare both monthly sales tables as one row per order line with compatible columns. Append them as new.
- 2 Declare Product Master as one row per ProductID. Test uniqueness before merge.
- 3 Merge sales (left) to product master (right) on ProductID using Left Outer when every sales row must remain visible.
- 4 Before expanding, inspect match behavior. Expand only needed attributes. Create a separate Left Anti query to list unmatched ProductIDs.
- 5 Reconcile sales row count and additive Net total before/after merge. If row count increases, inspect duplicate right-side keys before doing anything else.

EXPECTED RESULT

Append increases rows by the sum of source periods; merge enriches rows without multiplying the fact grain; exceptions stay visible.

WHY IT WORKED

The grain contract predicts what row count should do, turning unexpected multiplication into an immediate failure signal.

AX8 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: append two ticket months and merge Team metadata. The dirty Team master includes a duplicate TeamID. Diagnose and repair without DISTINCT/remove-duplicates as a blind fix.

COMMON BEGINNER MISTAKES

1. Appending a dimension to a fact table. | 2. Merging on a descriptive name instead of a stable key without justification. | 3. Expanding a many-match table and accepting row multiplication. | 4. Using Remove Duplicates after a bad merge to hide the symptom.

STUCK? RECOVERY

Write left grain, right grain and expected output grain. Check right-key uniqueness. Run left anti. Compare before/after row count and total. Only then revise the join. If still stuck, use only the partial skeleton in the practice workbook; save the attempt before opening review.

VALIDATE

Append row arithmetic correct; right key unique; unmatched list explained; fact row count and total stable after merge.

AX8 - Prove mastery

Page D - professional version, teach-back, review/retry and evidence

THEN SHOW THE PROFESSIONAL VERSION

Power Query merges should carry the same controls as SQL joins and BI relationships: key uniqueness, cardinality expectation, unmatched handling and before/after reconciliation.

TEACH IT BACK

Explain append vs merge using 'more rows' versus 'more columns'—then explain why that shortcut is not enough without grain.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry uses customer master with one missing key and a different join kind decision.

EVIDENCE / MASTERY

Grain contract + append/merge queries + anti-join exception + reconciliation + explain-back.

RECOMMENDED VIDEO: Power Query - Beginner to PRO Masterclass in 30 minutes

Visual editor support only; precise merge/append semantics use Microsoft references.

<https://www.youtube.com/watch?v=MMdcczmULrU>

AX9 - Parameters and reusable query design

Page A - why, outcome, prerequisite, mental model and vocabulary



WHY IT MATTERS

A file path, reporting date or environment value copied into several queries is easy to change inconsistently. A parameter centralizes configuration—but it is not a place to hide secrets or business definitions.

WHAT YOU LEARN

Create a parameter contract; define purpose/type/current value/allowed values/owner; reference it in a query; change one value and prove only the intended data changes.

PREREQUISITE

AX6 query basics. Understand that a parameter is configuration, not source data.

NEW WORDS BEFORE WE USE THEM

Parameter: A named value that can be reused in query logic or data-source functions. Configuration: A controlled setting that changes how the process runs without rewriting the core logic. Allowed value: A permitted parameter value or range used to reduce accidental misuse.

AX9 - Follow with me once

Page B - pStartDate controls the reporting window

pStartDate controls the reporting window

- 1 Create pStartDate as type Date and document why it exists, who changes it and what it should affect.
- 2 Reference pStartDate in the query filter instead of typing a literal date into several steps.
- 3 Before changing it, record current row count and total. Change only pStartDate, refresh, and predict direction of row-count change.
- 4 Compare before/after results. If unrelated fields change, inspect query dependencies.
- 5 Never store passwords/API secrets as ordinary worksheet parameters. Use appropriate credential mechanisms for real connections.

EXPECTED RESULT

One controlled configuration value changes the intended query behavior and leaves unrelated logic unchanged.

WHY IT WORKED

The parameter separates variable configuration from stable transformation logic and creates one auditable place to change the value.

AX9 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: create pRegion or pFolderPath with a documented contract and before/after validation.

COMMON BEGINNER MISTAKES

1. Calling a normal cell a parameter without connecting it to the query. | 2. Using parameters to hide a business rule that should be governed as data. | 3. Changing multiple parameters at once, making validation ambiguous. | 4. Storing credentials in plain workbook cells.

STUCK? RECOVERY

Reset to the previous value, refresh, record baseline controls, then change one parameter only. Trace every query step that references it. If still stuck, use only the partial skeleton in the practice workbook; save the attempt before opening review.

VALIDATE

Parameter purpose/type/owner recorded; only expected rows/files change; baseline can be restored reproducibly.

AX9 - Prove mastery

Page D - professional version, teach-back, review/retry and evidence

THEN SHOW THE PROFESSIONAL VERSION

Use parameters for controlled environmental or run-time values, combine them with clear query dependencies, and retain a release log when parameter changes affect published outputs.

TEACH IT BACK

Explain the difference between configuration, business data, and a secret.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry uses a different parameter type and expected effect.

EVIDENCE / MASTERY

Parameter contract + query reference + before/after controls + explain-back.

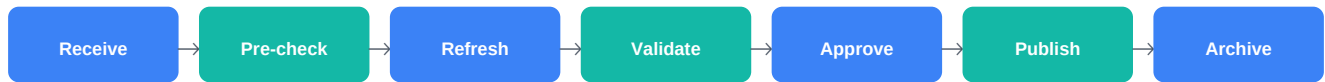
RECOMMENDED VIDEO: Power Query - Beginner to PRO Masterclass in 30 minutes

Editor context only; parameter contract is course-led.

<https://www.youtube.com/watch?v=MMdcczmULrU>

AX10 - Refreshable monthly reporting workflow

Page A - why, outcome, prerequisite, mental model and vocabulary



WHY IT MATTERS

A green Refresh All message only proves the engine ran. A professional monthly process must prove the expected files arrived, the correct scope was used, key errors were controlled, totals reconciled, and the release was approved.

WHAT YOU LEARN

Run a month-end release checklist: intake, expected/actual files, backup exclusion, parameter values, refresh, row/file/key/error controls, critical totals, issue resolution, approval and archive.

PREREQUISITE

AX1–AX9. This lesson integrates formula controls and Power Query controls.

NEW WORDS BEFORE WE USE THEM

Pre-check: Validation performed before refresh to confirm expected inputs and configuration. Release gate: A condition that must pass before a report is published. Rerun evidence: Saved proof that the process was executed again after a repair and that controls now pass.

AX10 - Follow with me once

Page B - April month-end refresh

April month-end refresh

- 1 Receive: place only expected source files in the intake folder or maintain a file whitelist. Record expected names and count.
- 2 Pre-check: confirm schema, date coverage, parameter values and backup exclusion before Refresh All.
- 3 Refresh: run queries. Save any error, not just the successful screen.
- 4 Validate: file contributions, total rows, duplicates, unmatched keys, error rows, critical Net total and Pivot/report reconciliation.
- 5 Approve: if all blocking controls pass, record release status and owner. If one critical control fails, do not publish.
- 6 Repair and rerun: change one cause, rerun the full relevant controls, then publish and archive source/version/evidence.

EXPECTED RESULT

A repeatable month-end run with a visible PASS/BLOCKED decision and evidence for every critical control.

WHY IT WORKED

The workflow treats refresh as one step inside a controlled release process rather than as the definition of success.

AX10 - Now do it yourself

Page C - fresh transfer, mistakes, recovery and validation

SMALL INDEPENDENT TASK

Fresh case: execute a May reporting cycle with one unexpected backup, one unmatched product and one corrected source file. Record block, repair and rerun evidence.

COMMON BEGINNER MISTAKES

1. Publishing because Refresh All showed no error. | 2. Leaving backup files in the source folder. | 3. Changing a parameter and forgetting to record it. | 4. Fixing a failed source file without rerunning reconciliations.

STUCK? RECOVERY

Return to the release checklist in order: expected inputs → configuration → query errors → row/key controls → totals → report outputs. Find the first failed control and repair from there. If still stuck, use only the partial skeleton in the practice workbook; save the attempt before opening review.

VALIDATE

Expected/actual files match; no uncontrolled duplicate/unmatched error; critical total reconciles; release owner/status saved; rerun evidence present.

AX10 - Prove mastery

Page D - professional version, teach-back, review/retry and evidence

THEN SHOW THE PROFESSIONAL VERSION

Treat recurring spreadsheet/Power Query reporting as an operational process with intake controls, release criteria, issue log, evidence archive and handoff—not as a personal workbook ritual.

TEACH IT BACK

Explain why 'refresh succeeded' and 'report is safe to publish' are different statements.

PROTECTED REVIEW + CHANGED RETRY

Attempt first. Changed retry uses a quarter-end folder with one schema drift and one late file.

EVIDENCE / MASTERY

Completed release checklist + error/recovery evidence + reconciliations + 90-second release defense.

RECOMMENDED VIDEO: Power Query - Beginner to PRO Masterclass in 30 minutes

33:35–35:17 refresh/load visualization; course book controls release discipline.

<https://www.youtube.com/watch?v=MMdcczmULrU>

C2-01 mastery checklist + quick reference

Use after the first study pass; this is not a substitute for practice

TEN PROFESSIONAL QUESTIONS

AX1 Where is each business assumption stored? | AX2 Is the lookup key clean/unique and what happens on no-match? | AX3 What should spill and how many rows should return? | AX4 Are numerator, denominator and period criteria explicit? | AX5 What question/aggregation does the Pivot answer and is it refreshed? | AX6 Which Applied Step first changes the result incorrectly? | AX7 Which files contributed and did schema drift occur? | AX8 What is input/output grain and should rows change? | AX9 What configuration does the parameter control? | AX10 What evidence allows release?

BLOCK RELEASE IF

Unexpected file scope; duplicate business key that changes enrichment; unexplained row multiplication; unresolved unmatched critical key; material reconciliation mismatch; query error removed without cause; stale Pivot/report; unsupported Excel feature being claimed as executed.

FAST REVISION

Formula layer: inputs → assumptions → calculations → outputs → checks. Lookup: key → uniqueness → exact match → exceptions. Power Query: source → types → transform → output → controls. Monthly run: intake → pre-check → refresh → validate → approve → publish → archive.

COURSE POSITIONING

This is optional Senior/BI depth. C2B may revisit ingestion, validation and grain with stronger engineering transfer. A learner choosing the direct DA→DE path can skip this C2A module after Course 1 and enter C2B when its prerequisites are met.

C2-01 source / video registry

Verified support; internal book remains sufficient if links fail

AX1/AX3/AX4 — RECOMMENDED

Chandoo - Best Excel Formulas in an hour

dVJFyqI_Nw4

Current title/timestamps resolved 2026-09-10.

AX2 — REQUIRED VISUAL SUPPORT

Chandoo - 10 Advanced XLOOKUP Tips & Tricks

YyhJe5tMq58

Current title/timestamps resolved 2026-09-10.

AX5 — REQUIRED VISUAL SUPPORT

Chandoo - Excel Pivot Tables - Master Data Analysis in just 45 minutes

zuSNd1ZMfBI

Current title/timestamps resolved 2026-09-10.

AX6-AX10 — REQUIRED/RECOMMENDED by lesson

Chandoo - Power Query - Beginner to PRO Masterclass in 30 minutes

MMdccczmULrU

Current title/timestamps resolved 2026-09-10.

AX2 — CURRENT REFERENCE

Microsoft Support - XLOOKUP

support.microsoft.com/en-us/excel/functions/xlookup-function

Reachable 2026-09-10; version support noted.

AX3 — CURRENT REFERENCES

Microsoft Support - UNIQUE / SORT

support.microsoft.com

Both current function pages reachable 2026-09-10.

AX6/AX9/AX10 — CURRENT REFERENCE

Microsoft Learn - Power Query best practices

learn.microsoft.com/en-us/power-query/best-practices

Updated 2026-08-20; reachable 2026-09-10.

AX7 — CURRENT REFERENCE

Microsoft Learn - Combine CSV files

learn.microsoft.com/en-us/power-query/combine-files-csv

Current page reachable 2026-09-10.

AX8 — CURRENT REFERENCES

Microsoft Learn - Append queries / Merge queries overview

learn.microsoft.com/en-us/power-query/

Current pages reachable 2026-09-10.

NO HIDDEN EXTERNAL PREREQUISITE

External sources visualize interfaces or reinforce current syntax. Every assessed business control—grain, key quality, source scope, reconciliation, failure recovery and release judgment—is taught inside the course and must be demonstrated in the learner's own files.